

Yujuan Zhang

Research focus: Bioinformatics, multi-omics, Machine Learning (ML), BioAI

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Appointments

Professor, College of Life Science, Chongqing Normal University (CQNU), China

Dec 2017 - Jan 2026 (completed)

Associate Professor, CQNU, China

Dec 2012 - Nov 2017

Assistant Professor, CQNU, China

Jul 2010 - Nov 2012

Education

PhD, Cell Biology, Chinese Academy of Sciences, China

Sep 2004 - Jul 2010

BS, Life Science, College of Life Sciences, Shaanxi Normal University, China

Sep 2000 - Jul 2004

Research Interests

- Multi-omics, Genomics, Cancer Biomarker, Precision diagnosis
- Protein language model-based functional and mechanistic inference
- Drug-protein-disease association, Drug repositioning

Technical Skills

- **End-to-End Data/AI (E2E):** data ingestion (ETL/ELT, Spark); SQL database (PostgreSQL); data modeling & metrics (dbt); workflow orchestration (Airflow); MLOps (containerization with Docker/Compose; experiment tracking & model registry with MLflow; CI/CD with GitHub Actions; deployment on AWS ECS/Fargate; monitoring with CloudWatch); web app/dashboard (Streamlit); data & infra on AWS (S3, RDS)
- **Programming:** Python, R; Unix/Linux; Shell
- **Machine Learning / Deep Learning:** scikit-learn, PyTorch; Transformers, PLMs, GNNs, VAEs
- **Data Analysis & Visualization:** Pandas, NumPy; data cleaning & EDA; Matplotlib; Jupyter
- **Statistics & Experimentation:** hypothesis testing, confidence intervals, A/B testing, experimental design
- **Interpretability:** SHAP/feature attribution; embedding analysis; attention/attribution analysis

- **Bioinformatics:** Multi-omics; NGS (WGS, WES, RNA-seq, scRNA-seq); variant calling; pathway/network analysis; NCBI/DrugBank/PubChem mining; PyMOL visualization

Teaching & Mentoring

Bioinformatics, Genomics and Proteomics

Selected Projects

LUAD Precision Oncology Platform (<https://luad-platform.streamlit.app/>)

- Designed and implemented a 7-module multi-omics workflow for TCGA-LUAD cohort analysis, encompassing somatic mutation profiling (MAF), bulk RNA-seq quantification, single-cell transcriptomics, ESM-based protein mutation encoder, and drug sensitivity mapping (OncoKB).
- Developed an end-to-end clinical report web application, deployed on cloud infrastructure.

Interpretable Protein LLMs for Ubiquitination prediction

- Built an ESM2-centered pipeline linking representation extraction, dimensionality reduction, and prediction.
- Developed interpretable models using embedding-based interpretation and attention-based analysis; contributed to code, data processing, and manuscripts.

Graph AI for biomedical link prediction (Drug Repositioning).

- Built a multi-view heterogeneous graph neural networks (GNN) to integrate noisy similarity networks and meta-paths for drug-protein-disease association prediction.
- Led implementation, benchmarking, and manuscript preparation.

Selected Publications (* Corresponding author)

1. Jinzhou Wu, Donglin He, Xin Li, Rui Wang, **Yu-Juan Zhang***. A multi-view collaborative heterogeneous graph neural network with semantic- and relation-aware for drug-disease association prediction. *Engineering Applications of Artificial Intelligence*. 2026. DOI: 10.1016/j.engappai.2025.113217.
2. Yawen Sun, Rui Wang, Zeyu Luo, Lejia Tan, Junhao Liu, Ruimeng Li, Dongqing Wei, **Yu-Juan Zhang***. ESM2_AMP: An Interpretable Framework for Protein-Protein Interaction Prediction and Biological Mechanism Discovery. *Briefings in Bioinformatics*. 2025. DOI: 10.1093/bib/bbaf434.
3. Zeyu Luo, Rui Wang, Yawen Sun, Junhao Liu, Zongqing Chen, **Yu-Juan Zhang***. Interpretable Feature Extraction and Dimensionality Reduction in ESM2 for Protein Localization Prediction. *Briefings in Bioinformatics*. 2024. DOI: 10.1093/bib/bbad534.
4. Junhao Liu, Zeyu Luo, Rui Wang, Xin Li, Yawen Sun, Zongqing Chen, **Yu-Juan Zhang***. EUP: Enhanced Cross-species Prediction of Ubiquitination Sites via a Conditional Variational Autoencoder Network Based on ESM2. *PLOS Computational Biology*. 2025 Jul 16;21(7):e1013268. DOI: 10.1371/journal.pcbi.1013268.
5. **Yu-Juan Zhang**, Zeyu Luo, Yawen Sun, Junhao Liu, Zongqing Chen*. From Beasts to Bytes: Revolutionizing Zoological Research with Artificial Intelligence. *Zoological Research*. 2023;44(6):1115-1131.

6. **Yu-Juan Zhang**, Yang Lan, Bin Chen*. ASDB: A comprehensive omics database for Anopheles sinensis. *Genomics*. 2021;113(3):976-982.
7. Xiaoyan Zuo, Bo Li, Chengxu Zhu, Zhengwen Yan, Li M, Xinyi Wang, **Yu-Juan Zhang***. Stoichiogenomics reveal oxygen usage bias, key proteins and pathways associated with stomach cancer. *Scientific Reports*. 2019 Aug 5;9(1):11344.
8. **Yu-Juan Zhang**, Chengxu Zhu, Yiran Ding, Zhengwen Yan, GongHua, Yang Lan, Jian-Fan Wen*, Bin Chen*. Subcellular stoichiogenomics reveal cell evolution and electrostatic interaction mechanisms in cytoskeleton. *BMC Genomics*. 2018;19(1):469.

Books

- **Yu-Juan Zhang**, Boqi Li, Baoli Qiu. Applications of Artificial Intelligence in Life Sciences. Science Press, 2026. (In press).
- **Yu-Juan Zhang** and Wen JF. Proteobacteria and the endosymbiotic origin of mitochondrion. In: Maria L. Sezenna (ed.). Proteobacteria: Phylogeny, Metabolic Diversity and Ecological Effects. Nova Science Publishers, 2011, pp. 57-71.

Patents and Software Copyrights

- Intelligent Assisted Ubiquitinated Protein Detection Platform V1.0 (Software Copyright). 2025. Registration No. 2025SR0582782. National Copyright Administration of the PRC.
- A Computational Method for Basic Stoichiometric Genomic Analysis (Invention Patent). Application No. 202110905333.9.
- A Joint Analysis Method for Stoichiometric Genomes and Genome Annotation Files (Invention Patent). Application No. 202110906004.6.
- A Batch Analysis Method for Multi-species Stoichiometric Genomes (Invention Patent). Application No. 2021109053.
- A High-throughput Method for Multi-species Stoichiometric Proteome Analysis (Invention Patent). Application No. 202210259556X.
- A High-throughput Method for Multi-species Stoichiometric Transcriptome Analysis (Invention Patent). Application No. 202210159540.

Certifications

- Machine Learning Scientist in Python, Dec 2025, DataCamp.
- Associate Data Scientist in Python, Nov 2025, DataCamp.